1. Introduction

The goal of this assignment is to determine how many amino-acid substitutions a protein can handle before its structure is affected. To do this, we needed to start with a high-quality experimental structure, with good geometry, refinement statistics, and no chain breaks, so that any structural changes observed later could be attributed to the introduced mutations rather than limitations of the original model.

The mutation strategy combined basic principles from protein evolution and structural biology. In nature, amino-acid substitutions do not occur randomly, but they follow evolutionary patterns captured by matrices like BLOSUM62 (Henikoff & Henikoff, 1992). In addition, the structural environment of each residue plays a big role in how easily it can change. Because modifications to the protein core are more likely to upset stability, buried residues typically accept conservative replacements, whereas exposed residues are typically more flexible (Tokurikio & Tawfik, 2009). In order to prevent substitutions that could result in steric conflicts or cavities, I additionally took side chain volume (IMGT data) into account for the buried locations (Tsai et al., 1999). Using these constraints, I generated mutated versions of the 6E6O sequence, reflecting both evolutionary likelihood and basic structural constraints.

The Cα root mean square deviation (RMSD), a common metric for evaluating fold preservation following optimal superposition, was used to quantify the structural similarity between the natural protein and the mutant models (Zhang et al., 2004). It is defined as:

$$RMSD = \sqrt{\frac{1}{N}\sum_{i=1}^{N} || \mathrm{x_i} - \mathrm{y_i} ||^2}$$

where N is the number of aligned Cα atoms and x_i, y_i are their Cartesian coordinates after superposition.

2. Methods

2.1 Selection and quality assessment of the protein structure

To select a suitable protein, I searched the PDB (RCSB PDB: Homepage) for small, high-quality X-ray structures and excluded lysozyme as required. I focused on proteins of ~50 amino acids to keep the computational cost low, and prioritized entries with high resolution, low R_work/R_free values and generally favourable validation metrics. For the final choice, I examined the full wwPDB validation report, looking at resolution, Ramachandran and rotamer statistics, clashscore, and bond-geometry RMSZ and RSRZ values to ensure that the structure was well refined and free of modelling issues.

2.2 Verification of chain integrity using Bio.PDB

I loaded the PDB file using Bio.PDB and measured the lengths between successive Cα atoms in chain A to confirm that the 6E6O structure contained no chain breaks. I applied a 4.5 Å threshold to identify possible discontinuities because the Cα–Cα gap in a continuous polypeptide backbone is approximately 3.8 Å (Dal Palù et al., 2011). All distances fell below this cutoff, confirming that the chain is fully intact.

2.3 Introduction of mutations, using a systematic computational strategy

I created a repeatable Python script utilizing Bio.PDB to produce mutations in a regulated and physiologically justified manner. For each residue, I estimated how buried it is by counting nearby Cα atoms within 10 Å, and classified positions as buried or exposed. Substitutions were then sampled from the BLOSUM62 matrix: exposed residues could mutate according to the full distribution, while buried residues were restricted to conservative substitutions and additionally filtered by side-chain volume (IMGT data) to avoid steric clashes or cavities. Since changes larger than ~30–50 $Å^3$ tend to destabilize the protein core, I used a 40 $Å^3$ threshold (Tsai et al., 1999; IMGT, n.d.). The BLOSUM log-odds scores were converted into probability distributions so that more likely substitutions were sampled more frequently. A fixed random seed ensured reproducibility. Using this procedure, I generated mutant sequences at different mutation levels, and for each level I produced three independent variants using seeds 3, 4 and 6 to test whether the results depended on the specific mutation pattern.

2.4 Structural assessment of mutant models

To assess whether the native fold was preserved after mutation, I used the Cα RMSD between each predicted mutant structure (generated with AlphaFold2) and the native 6E6O structure. Superpositions were performed in PyMOL using the *align* command, which reports the final Cα RMSD after outlier removal. I defined RMSD ≤ 3 Å as preservation of the fold and RMSD ≥ 6 Å as clear disruption. These thresholds reflect typical same-fold comparisons, where Cα RMSD values below ~3 Å indicate fold preservation, while values above ~6 Å correspond to major topological rearrangements. I defined these RMSD thresholds a priori and used them consistently across all mutation levels and seeds to evaluate structural similarity.

2.5 Deep-learning structure prediction using AlphaFold2

To predict the 3D structures of all mutant sequences, I used AlphaFold2, because it provides accurate backbone geometries, even for sequences with many substitutions, and therefore allows reliable RMSD-based comparison to the native fold. AlphaFold2 predicts structure directly from the amino-acid sequence using two main components: the Evoformer, which models long-range residue interactions through attention mechanisms, and the Structure Module, which converts these geometric predictions into 3D coordinates. The Frame Aligned Point Error (FAPE), which measures how closely the predicted local frames match the actual atomic positions, is used to train the model. To guarantee similar structural predictions, this pipeline was applied uniformly to every mutant. (Where: R_k and t_k are the predicted rotation and translation for residue k and x_i^{true} is the true atom position (Jumper et al., 2021)).

$$\text{FAPE} = \frac{1}{N}\sum_{i=1}^{N} \left| \ \text{clip}(\| R_k x_i + t_k - x_i^{\text{true}} \|) \ \right|$$

AlphaFold2 was used to check whether the native fold was preserved, rather than to evaluate thermodynamic stability. For small single-domain proteins, it can still produce

reliable backbone geometries even without MSAs, so comparing the predicted structures across mutants is more informative than looking at the absolute confidence scores.

2.6 Assessment of prediction reliability

To evaluate the reliability of the predicted structures, I combined AlphaFold2's internal confidence scores with structural similarity measures. Model confidence was evaluated using the per-residue pLDDT values, which show how confident the model is about the local geometry of each residue. High pLDDT values indicate well-defined regions, whereas low values correspond to uncertain or flexible parts of the structure. I also quantified structural similarity to the native fold using the Cα RMSD after optimal superposition, computed with BioPython's *Superimposer*.

3. Results

3.1 Selection and Evaluation of the Protein Structure

I selected PDB entry 6E6O, a 40-residue pheromone from Euplotes raikovi, which meets all criteria defined in the Methods. The wwPDB validation report confirms that it is a high-quality X-ray structure. The resolution is exceptionally high (0.70 Å), and the refinement statistics (R = 0.165, R_free = 0.184) indicate excellent agreement with the experimental data. There are no Ramachandran or rotamer outliers, and the clashscore is low (5). The bond length and bond angle RMSZ values (1.23 and 1.20) are nearly ideal, while prolines have a few high RSRZ values, which makes sense given their ring restrictions. Overall, the structure shows very good geometry and a strong fit to the electron density, making 6E6O a suitable and reliable starting model for this assignment.

3.2 RMSD analysis across mutation levels and seeds

For each mutation level (5–40%), three independently generated mutant sequences (seeds 3, 4 and 6) were predicted and aligned to the native 6E6O structure. Seed 3 showed consistently low RMSD values across all mutation levels (~2.7–3.1 Å), indicating preservation of the fold even at 40% mutations. Seed 4 behaved very differently: RMSD increased sharply at 15% (~6 Å) and remained high at all higher mutation levels, showing clear loss of the native topology. Seed 6 remained stable up to 30% mutations (~2.7–3.1 Å) but collapsed at 40% (RMSD ~6.6 Å) (Table 1, Figure 1).

Overall, these results show that mutation tolerance in 6E6O depends strongly on the specific substitution pattern rather than the mutation percentage itself. Some mutation patterns preserve the fold even at high substitution levels, while others disrupt it much earlier.

Table 1. Cα RMSD values and Mean pLDDT for mutant structures

Mutation level	**5%**	**10%**	**15%**	**20%**	**25%**	**30%**	**40%**	
RMSD	2.993	3.068	2.780	2.837	2.727	2.841	3.023	Seed 3
	2.734	3.051	6.073	6.090	6.171	6.473	6.566	Seed 4
	2.768	2.868	2.868	2.724	2.918	3.063	6.561	Seed 6
Mean pLDDT	70.4	65.5	64.3	65.3	69.4	66.7	68.2	Seed 3
	64.7	68.2	68.4	67.9	67.1	66.7	62.6	Seed 4
	63.4	69.7	69.7	67.6	66.0	68.1	64.5	Seed 6

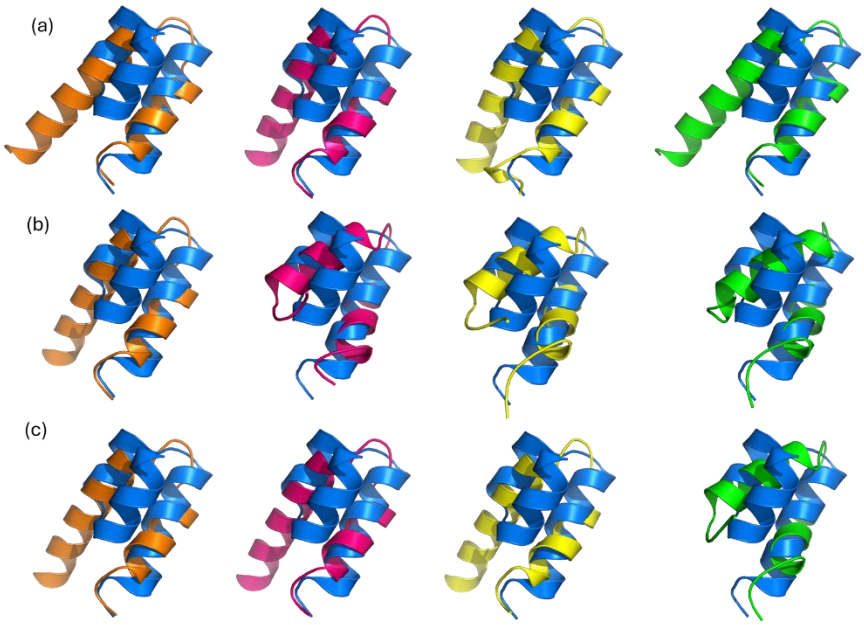

Figure 1. Representative superpositions of mutant models onto the native 6E6O (blue) visualized in PyMOL. (a) seed = 3, (b) seed = 4, (c) seed = 6 . Orange: 5% mutations, pink: 15% mutations, yellow: 30% mutations, green: 40% mutations.

3.3 Mean pLDDT scores

For every mutant sequence, the average pLDDT was calculated per Cα atom, with higher values indicating greater confidence in local geometry (Table 1). Overall, the predicted local structures have moderate to high confidence, as indicated by the mean pLDDT values, which range from about 62 to 70. pLDDT reflects local residue confidence, not whether the overall fold is correct, so even misfolded models can score well. For assessing fold retention, Cα RMSD after superposition, especially when consistent across independent mutants, is a more reliable indicator. The majority of residues are still rather well predicted, even at greater mutation levels.

4. Discussion

The results show that the structural tolerance of 6E6O depends more on the specific substitution pattern than on the overall mutation percentage. Seeds 3 and 6 preserved the native fold even at high mutation levels, whereas seed 4 collapsed already at 15%, likely due to substitutions in buried or structurally sensitive positions. For a small and tightly packed protein like 6E6O, a few incompatible changes are enough to destabilize the fold. RMSD proved to be a clear and reliable indicator of fold preservation, with the predefined thresholds matching the visual superpositions in PyMOL. AlphaFold2 predicted all mutants with moderate but consistent confidence (mean pLDDT ≈ 62–70), but pLDDT alone could not distinguish stable from unstable mutants. RMSD and the agreement across seeds provided a more informative assessment of structural stability. Although AlphaFold2 without MSAs has limited confidence and the mutation strategy cannot capture epistatic effects, the combined use of RMSD, pLDDT and seed-to-seed consistency gave a coherent picture of when the structure remained intact and when it broke down. Overall, the study demonstrates that fold stability in 6E6O is very sequence

dependent, and that significant mutation can be tolerated as long as essential structural sites are preserved. This approach has limitations because it only uses in silico predictions from one crystal structure and doesn't test them in the lab or consider different conformations or environmental factors. In addition, burial was estimated using Cα-neighbor counts within 10 Å, which is a simplified approximation that may misclassify borderline residues.

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